CMSC320 Introduction to Data Science

Writing Assignment 4

In this assignment, we'll study a visualization and create a visualization.

Part 1. Studying a visualization and making sense of it and its underlying data.

Have you taken the DC/MD/VA subway? Taking it before you do this part may give you more domain knowledge and make your work easier/more meaningful to you. The College Park station is a 20-minute walk from campus and accessible by bus too.

1.1. Read this Reddit post: https://www.reddit.com/r/washingtondc/comments/15mbos4/i_mapped_the_layouts_of_all_98_metro_stations_so/

1.2. The first GitHub link in the Reddit post works and is the main source of visualizations and data for this part of the assignment: https://github.com/eable2/DCMetroStationExits/blob/main/WMATA%20Metro%20Station%20Platform%20Exit%20Guide%20June%202025.pdf

1.3. You can find the different files on the left-hand side. You can see the visualization in the main file (name starts with WMATA) by clicking the file name, scrolling down and clicking "More pages." You can download this pdf file by clicking a download button in the upper-right corner of the webpage.

1.4. Think about how you make sense of the images in the pdf and what you use to do that (the write-up at the top, the "How to Read a Station Diagram" page, the data files?).

1.5. Comment on the data: how it's split into files and why that may be; how it might have been collected.

1.6. Comment on how the visualizations may have been created (Photoshop? Something more automated/programmatic?).

1.7. Comment on whether this data viz work is useful to you and whether you think it may be useful for society at large (Reddit comments provide a glimpse into society's feedback).

1.8. In what specific context might this be useful to someone? Would all images be useful to one person or maybe just a small number?

1.9. What questions would you ask the creator? Have they maybe answered that on Reddit?

Part 2. Learning about a visualization type, learning to create an instance of that, thinking of what to visualize and creating a visualization.

2.1. Learn what a Sankey diagram is: https://en.wikipedia.org/wiki/Sankey_diagram. Feel free to Google around for more examples.

2.2. Use this tool to create a Sankey diagram: https://sankeymatic.com/build/. This webpage, on the left, has controls that make things happen. The website also has step-by-step examples, for example: https://sankeymatic.com/gallery/monthly-budget.html. Play/look around to learn how things work.

2.3. Decide on something to visualize. It can be something from a dataset that you find, or something from your life (like your grad school application process).

2.4. Make a diagram and share it with the reader. Use accompanying text if helpful.

Submission Instructions:

Create a single pdf file where you address the questions/tasks above. Structure it however you wish, assuming that your reader can't read your mind. For example, you should write about what data you chose to visualize and why and how you procured it. The doc will be read from beginning to end and the reader will grade you on how well they understand your thought process and workflow.

You will be graded on **effort**, but **leniently**.

If your total word count is under 1000, consider fleshing out some of your thoughts. Over 2000 words is probably too many.

See the rubric at the bottom of the Canvas assignment page for the rubric that will be used to grade your work.

Submit the pdf file to Canvas.
Name the file as follows: 'CMSC320_WA3_YourName.pdf'

REMEMBER: The use of AI tools in generating responses is not allowed.